10. Key Business Insights

This chapter consolidates the findings from exploratory data analysis, correlation analysis, machine learning modelling, SHAP interpretation, and interactive dashboard analysis to identify the key factors influencing Airbnb market performance and investment opportunities.

The analysis reveals that Airbnb success is primarily driven by three major factors: **location attractiveness, property characteristics, and demand intensity**. While host-related factors and amenities contribute to customer experience, they have a relatively smaller influence on pricing and revenue outcomes compared with geographical and property-level attributes.

Insight 1 — Location is the primary driver of airbnb performance

The strongest insight from both exploratory analysis and machine learning interpretation is that **location is the most important determinant of Airbnb pricing and investment potential**.

Correlation analysis showed that geographical variables, particularly latitude and longitude, had meaningful relationships with listing prices. The machine learning model further confirmed this finding through SHAP analysis, where **latitude was ranked as the most influential feature**, with the largest contribution to predicted prices.

The SHAP results demonstrate that location does not follow a simple linear relationship with price. Instead, certain geographical areas create significant price premiums due to factors such as tourist attraction proximity, accessibility and transportation connections, neighbourhood reputation, local tourism demand, and availability of competing listings. This indicates that investors should avoid evaluating properties based only on city-level performance. Within each city, specific neighbourhoods generate substantially different revenue opportunities. For example, dashboard analysis identified that central and highly visited neighbourhoods consistently achieved stronger demand and pricing performance compared with peripheral areas. Therefore, **neighbourhood-level selection is more important than simply selecting a high-performing city**.

Insight 2 — Property size and capacity strongly influence revenue

The machine learning models consistently identified property characteristics as major contributors to Airbnb price prediction. Among all property-related variables, bedrooms and accommodate were the strongest predictors after location.

The SHAP analysis showed a clear positive relationship, indicating that listings with fewer bedrooms generally reduced predicted prices, and larger properties increased predicted prices due to higher accommodation capacity. This finding aligns with customer behaviour because larger properties are often preferred by families, group travellers, business groups, and longer-stay customers. The correlation analysis also supported this relationship, showing positive associations between property size variables and listing prices. However, larger properties require higher initial investment and operational costs. That also means a medium-to-large property in a high-demand neighbourhood may generate better returns than a luxury property in a low-demand location. Therefore, investors should consider revenue potential rather than only selecting the most expensive property category.

Insight 3 — Demand analysis reveals the importance of market activity

The demand analysis dashboard used review activity as a proxy for Airbnb customer demand because booking transaction data was unavailable. The analysis of millions of review records revealed clear differences in market activity between cities and neighbourhoods. It can be seen that the demand is highly concentrated in specific locations. The result also shows that a small number of neighbourhoods generate a large proportion of total customer activity. Besides, Popular areas maintain stronger and more consistent monthly demand trends.

The demand dashboard also highlighted seasonal patterns, showing that Airbnb markets experience fluctuations throughout the year. Peak periods create opportunities for higher pricing strategies, while low seasons require competitive pricing and improved occupancy strategies. In addition, a location with consistently strong demand may provide more reliable returns than an area with occasional high-price opportunities.

Insight 4 — Host characteristics and amenities have insignificant influence

Although host-related variables were included in the analysis, their impact was considerably smaller compared with location and property characteristics. Features such as host acceptance rate, host response rate, host identity verification, instant booking availability, and host experience showed limited influence in the SHAP analysis.

Similarly, amenity count had only a moderate effect on predicted prices.

This suggests that while good management practices improve customer satisfaction and operational performance, they cannot compensate for poor location or unsuitable property characteristics.